

In a certain FET circuit , $V_{GS} = 0 \text{ V}$, $V_{DD} = 15 \text{ V}$, $I_{DSS} = 15 \text{ mA}$, and $R_D = 470 \text{ }\Omega$. If R_D is decreased to $330 \text{ }\Omega$, I_{DSS} is

Select one:

- ☐ a. 10.5 mA
- ☐ b. 15 mA
- ☐ c. 19.5 mA

Which of the following is a false statement regarding the dc load line when comparing self-bias and voltage-divider configurations?

Select one:

- ☐ a. Both cross the origin.
- ☐ b. Both are linear lines.
- ☐ c. Both intersect the transfer characteristics.
- ☐ d. Both are obtained by writing Kirchhoff's voltage law (KVL) at the

A certain P-channel E-MOSFET has $V_{G_s(th)} = -2V$, If $V_{G_s} = 0V$, the drain current is:

Select one:

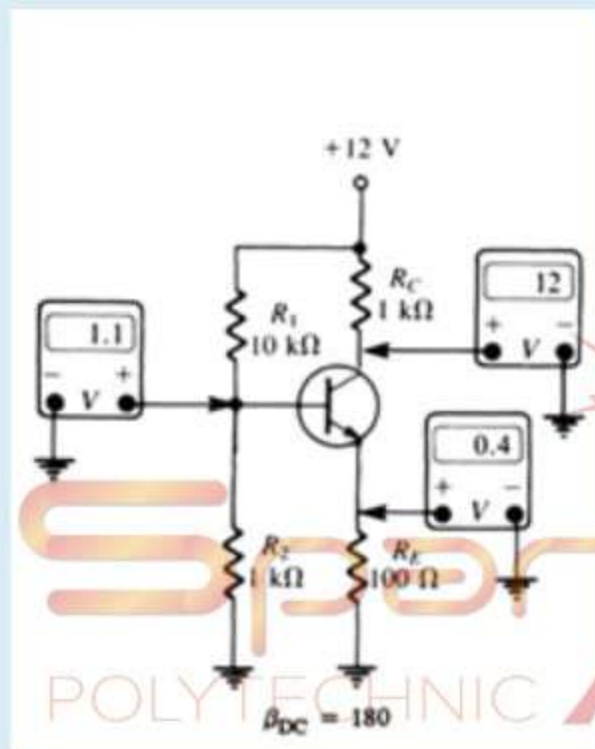
- ☒ a. maximum
- ☐ b. $I_{D(on)}$
- ☐ c. I_{DSS}
- ☐ d. 0 mA

What is the approximate current level in the gate of an FET in dc analysis?

Select one:

- ☐ a. 0.3 mA
- ☐ b. Undefined
- ☐ c. 0 A
- ☐ d. 0.7 mA

Refer to Figure shown. The most probable cause of trouble, if any, from these voltage measurements would be



Select one:

- ☐ a. the base-emitter junction is open.
- ☐ b. a short from collector to emitter
- ☐ c. R_E is open
- ☐ d. no problems

For the FET, the relationship between the input and output quantities is _____ due to the

_____ term in Shockley's equation.

Select one:

- ☐ a. nonlinear, cubed
- ☐ b. linear, proportional
- ☐ c. nonlinear, squared

AN n-channel D-MOSFET with a positive V_{GS} is operate in

Select one:

- ☐ a. the depletion mode
- ☐ b. the enhancement mode
- ☐ c. cutoff
- ☐ d. saturation

